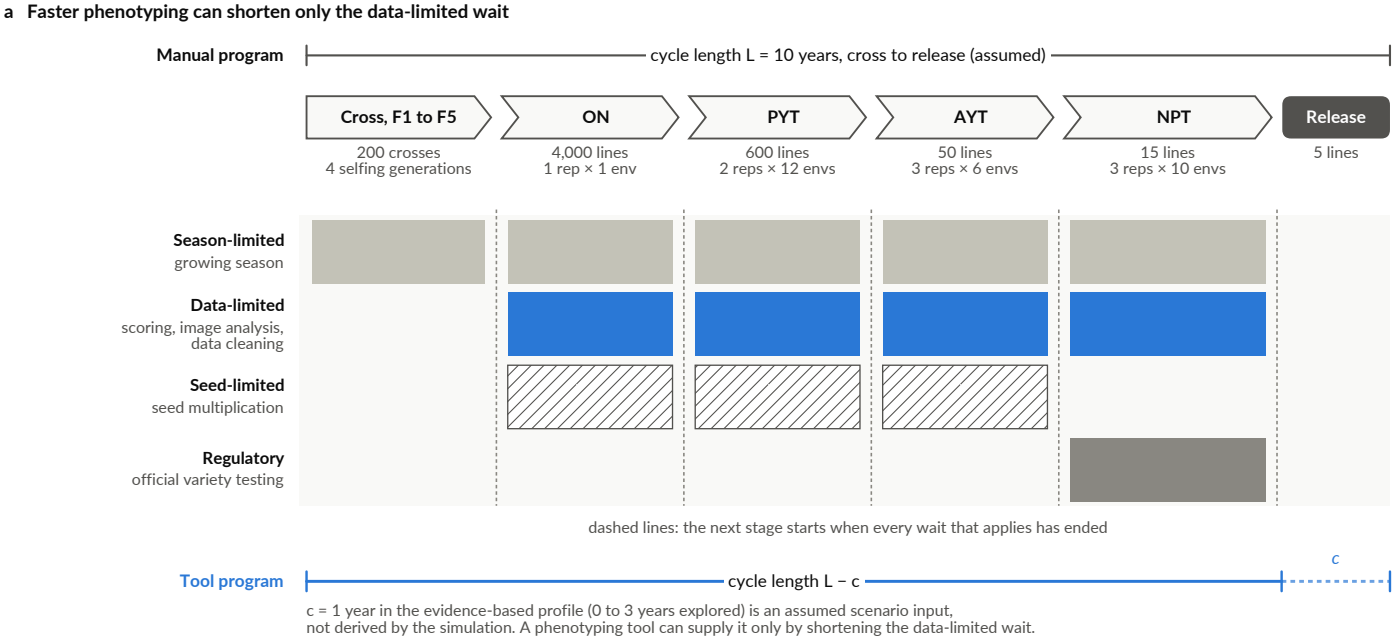
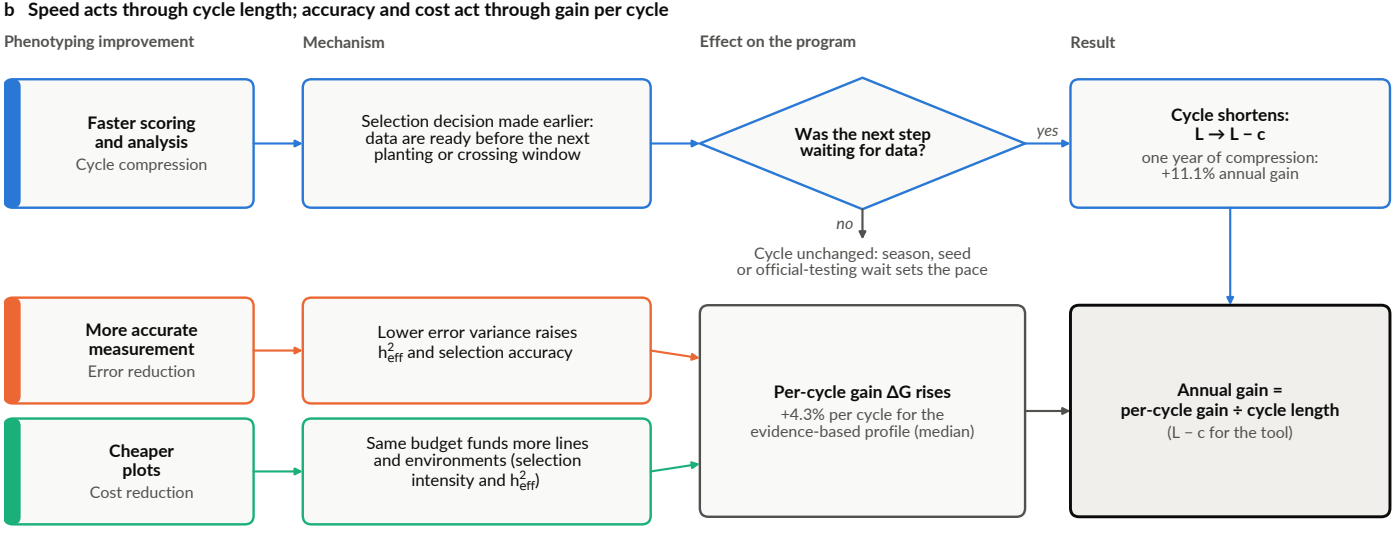

a Faster phenotyping can shorten only the data-limited wait
Manual program
cycle length L = 10 years, cross to release (assumed)
Cross, F1 to F5
200 crosses
4 selfing generations
ON
4,000 lines
1 rep × 1 env
PYT
600 lines
2 reps × 12 envs
AYT
50 lines
3 reps × 6 envs
NPT
15 lines
3 reps × 10 envs
Release
5 lines
Season-limited
growing season
Data-limited
scoring, image analysis, data cleaning
Seed-limited
seed multiplication
Regulatory
official variety testing
dashed lines: the next stage starts when every wait that applies has ended
Tool program
cycle length L − c
c
c = 1 year in the evidence-based profile (0 to 3 years explored) is an assumed scenario input,
not derived by the simulation. A phenotyping tool can supply it only by shortening the data-limited wait.


b Speed acts through cycle length; accuracy and cost act through gain per cycle
Phenotyping improvement
Mechanism
Effect on the program
Result
Faster scoring and analysis
Cycle compression
Selection decision made earlier: data are ready before the next planting or crossing window
Was the next step waiting for data?
yes
Cycle shortens: L → L − c
one year of compression: +11.1% annual gain
no
Cycle unchanged: season, seed or official-testing wait sets the pace
More accurate measurement
Error reduction
Lower error variance raises h^2_{eff} and selection accuracy
Cheaper plots
Cost reduction
Same budget funds more lines and environments (selection intensity and h^2_{eff})
Per-cycle gain ΔG rises
+4.3% per cycle for the evidence-based profile (median)
Annual gain = per-cycle gain ÷ cycle length
(L − c for the tool)